AI Accelerator - QA Track

Week 3 Participant Execution Guide

Continue using the same Membership App repository from Week 2. Week 3 focuses on test-failure root-cause analysis, AI-assisted testing, and regression prioritization with coverage-gap analysis.

Before You Start

Continue using the **same Membership App repository** you used for Week 2. You do not need to fork the repository again. Make sure your local repository is available and your Week 2 work is preserved before proceeding.

1. Update Your Local Repository and Checkout the Week 3 Branch

Open your local Membership App repository and fetch the latest branches:

```
git fetch --all
```

Checkout the Week 3 branch:

```
git checkout l3/week3
```

If the branch is only available from the remote:

```
git checkout -b l3/week3 origin/l3/week3
```

- Confirm that your current branch is **l3/week3**.
- Set up/run the Membership App locally.
- Confirm that the application is working before starting the activities.

> **Important:** Perform the Week 3 activities against the **l3/week3** version of the Membership App.

Activity 1 - Test-Failure Root-Cause Analysis

The goal is to investigate **actual failures/issues** and use AI to assist with Root Cause Analysis (RCA), while validating the AI's conclusions against real evidence.

2. Run and Explore the Week 3 Application

- Run the Membership App locally.
- Review the Week 3 application and execute relevant tests/scenarios.
- Identify **at least 5 actual issues/failures** to investigate.
- Use actual Week 3 application behavior; do not invent hypothetical failures only to complete the activity.

3. Investigate Each Failure

For each issue/failure:

- Reproduce the failure.
- Record the expected result.
- Record the actual result.

- Capture available evidence such as screenshots, error messages, logs, stack traces, test execution results, or relevant code.
- Use AI to help analyze the failure and determine the possible root cause.
- Save the **exact AI prompt and exact AI-generated output** in your Markdown (.md) evidence file.

4. Validate the AI's RCA

Do not accept the AI explanation automatically. Trace its claims back to actual evidence:

> **Application behavior -> Logs / Stack Trace -> Code -> Requirements**

For each failure, document:

- Issue/failure
- Steps to reproduce
- Expected result
- Actual result
- AI analysis
- Evidence reviewed
- Validated root cause
- Severity / impact
- Blast radius
- Recommended fix or next diagnostic step

> If the available evidence is not enough to determine the root cause, state that clearly instead of forcing a conclusion.

5. Complete Activity 1 in the QA Training Platform

- Use https://qa.mprestoza.com to complete the Week 3 RCA activity.
- Provide the required findings and supporting evidence.
- Prepare your RCA deliverable in the required format (.md or PDF, as applicable to the activity submission).

Activity 2 - AI-Assisted Test

Use AI to help create a test based on an actual feature/requirement, then validate and execute what AI produced.

6. Choose a Real Feature or Requirement

- Select a feature or requirement from the Membership App that is appropriate for testing.
- **Manual QA:** AI may assist with structured test scenarios, test steps, expected results, test data, or another appropriate QA test artifact.
- **Automation QA:** AI may generate or assist with an automated test using your chosen/available automation framework.

- The examples are guidance only. Choose an artifact that makes sense for your actual QA workflow.

7. Generate the Test With AI

- Provide AI with the relevant requirement and context.
- Save the exact prompt and exact AI output.
- Save any follow-up prompts and revised AI outputs.
- Do not silently correct the generated output before documenting it.

8. Validate and Execute the Test

Review the AI-generated test against:

- Actual requirement
- Actual application behavior
- Existing test conventions
- Framework conventions, when applicable
- Existing code/tests, when relevant

> **Document:** AI generated -> What you validated -> What you changed/rejected -> Final test -> Execution result -> Evidence

- **Automation QA:** Include actual test execution evidence.
- **Manual QA:** Include actual execution results and supporting evidence from the application.

9. Complete Activity 2 in the QA Training Platform

Submit/document the AI-assisted test and the required evidence through the QA Training Platform.

Activity 3 - Regression Prioritization & Coverage-Gap Analysis

Determine what should be regression tested and what testing coverage may be missing based on the actual Week 3 changes.

10. Review the Actual Week 3 Changes

- Use the actual Week 3 branch and available requirements/code changes as your source.
- Identify what changed.
- Identify which features/components may be affected.
- Review existing tests related to those areas.
- Identify potential regression risks.

11. Use AI to Assist With Regression Analysis

Provide AI with relevant context from the actual **changes/code -> requirements -> existing tests**.

- Ask AI to identify potentially impacted areas.
- Ask AI to prioritize regression tests.

- Ask AI to identify possible coverage gaps.
- Ask AI to suggest additional tests where appropriate.
- Save the exact prompts and AI outputs in your Markdown (.md) evidence file.

12. Validate the AI's Recommendations

Do not accept the regression list simply because AI generated it. Cross-check the recommendations against:

- Actual changed code/files
- Actual requirements
- Existing tests
- Application behavior
- Known dependencies/affected areas

> Document which recommendations you **Accepted -> Modified -> Rejected -> Added yourself**, and explain why. Your final regression priorities and coverage gaps should represent your validated QA judgment, not simply the AI response.

13. Complete Activity 3 in the QA Training Platform

Record your regression prioritization, coverage-gap findings, validation decisions, and supporting evidence in the Week 3 activity.

14. Complete Your Week 3 Daily Logs

Complete the required **5 Daily Logs for Week 3**. Your logs should reflect the actual work performed during the week.

15. Review Your Week 3 Deliverables

Before completing Week 3, verify that you have:

- Completed RCA for at least **5 actual issues/failures**.
- Validated AI RCA against actual evidence.
- Completed one AI-assisted test.
- Executed and validated the test.
- Completed regression prioritization.
- Completed coverage-gap analysis.
- Saved your exact AI prompts and AI outputs.
- Provided the required supporting evidence.
- Completed Activities 1-3 in qa.mprestoza.com.
- Completed all **5 Week 3 Daily Logs**.

16. Record Your Week 3 Weekly Summary Video

After completing the activities, record your **Week 3 weekly summary video**. Use your actual submissions and evidence while walking through your work.

- Walk through your RCA investigation and at least one meaningful failure/root cause.
- Explain how AI helped with the investigation.
- Show where you validated, corrected, or rejected AI's analysis.
- Show your AI-assisted test and its execution result.
- Explain your regression prioritization and coverage-gap analysis.
- Show evidence supporting your conclusions.
- Briefly explain what you learned from the Week 3 activities.
- You may use any screen-recording tool available to you.

> The recording should demonstrate your **actual work and QA reasoning**, rather than simply reading your submitted answers.

Final Week 3 Execution Flow

> **Open Existing Membership App Repo -> Fetch Latest Branches -> Checkout l3/week3 -> Run Application -> Activity 1: Investigate 5 Actual Failures + RCA -> Activity 2: Create, Validate & Execute AI-Assisted Test -> Activity 3: Regression Prioritization + Coverage-Gap Analysis -> Complete Evidence -> Complete 5 Daily Logs -> Record & Submit Weekly Summary Video**